The questions of this chapter

Given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, consider

$$Ax = b.$$

Separate four questions.

1. Does a solution exist?
2. Is the solution unique?
3. How can a finite procedure find every solution?
4. What are the arithmetic cost and numerical risks?

This chapter addresses the first three. Chapter 11 adds finite precision, stability, and condition numbers.

An absence of contradictory rows proves consistency. It does not by itself prove uniqueness.

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Notation and the underlying field

- m is the output dimension, number of equations, and number of rows of A.
- n is the input dimension, number of unknowns, and number of columns of A.
- A_{ij} is the entry in row i, column j.
- e_i is standard basis vector i.
- P denotes a permutation matrix.
- $\mathbb{R}$ or $\mathbb{C}$ is the field containing the entries.

If A is $m \times n$, then x must be $n \times 1$ and b must be $m \times 1$. Check these shapes before multiplying or adding.

Dimension errors can be detected before any numerical calculation.

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Two pictures of one system

Consider

$$x + 3y = 7, \qquad 2x - y = 0.$$

In the row picture, the equations are two lines in the xy plane. They intersect at $(x, y) = (1, 2)$.

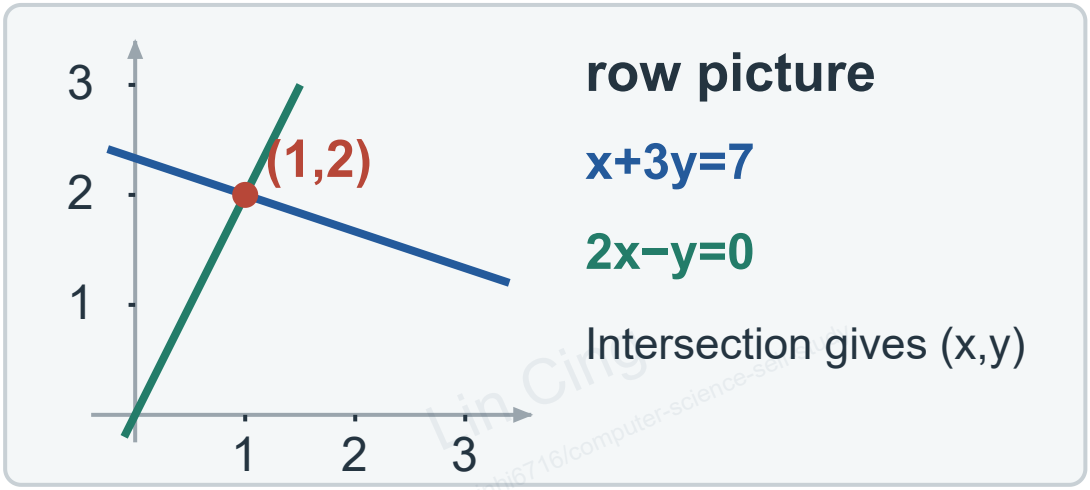
Column picture:

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}.$$

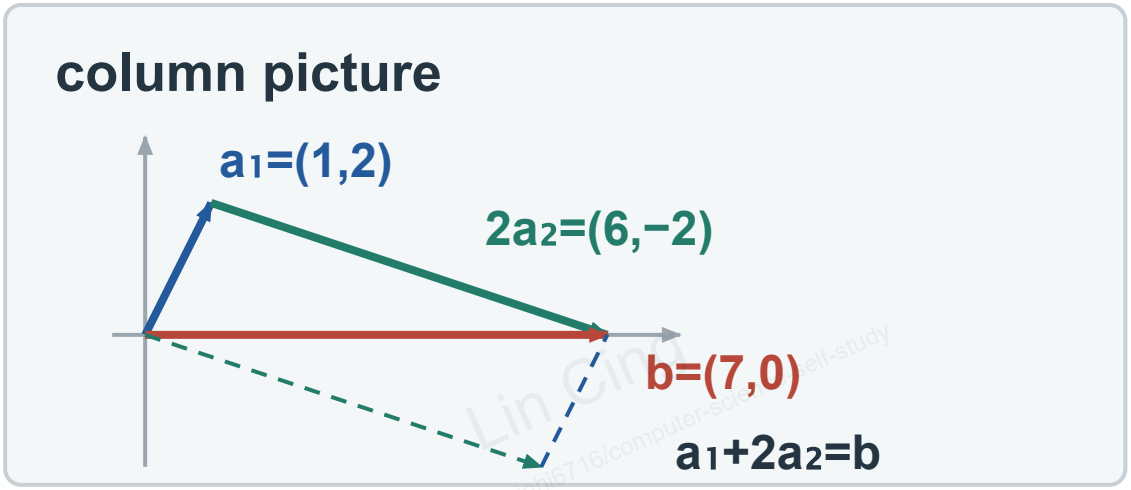
The coefficients $(1, 2)$ combine the columns into the target. Both pictures describe the same numbers. Rows express constraints, while columns express reachable directions.

In higher dimensions, the row picture uses planes or hyperplanes. The column picture still uses coefficients of vector combinations.

Row picture: intersection of the equations



Column picture: combining columns into the target



Existence and uniqueness

A real linear system can have the following outcomes.

- A unique solution if every variable has a pivot and no row is contradictory.
- Infinitely many solutions if the system is consistent and has a free variable.
- No solution if a row is $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$.

For $A \in \mathbb{R}^{m \times n}$,

- If $m < n$, the homogeneous system $Ax = 0$ has a nonzero solution, but $Ax = b$ may still be inconsistent.
- Every $b \in \mathbb{R}^m$ is attainable exactly when the column space is $\mathbb{R}^m$.
- Each consistent right-hand side has a unique solution exactly when the nullspace contains only zero.

Chapter 3 formalizes rank-nullity. For now, count free variables.

A complete consistency decision

Solve

$$\begin{aligned} x &+ 2y &&= 4, \\ 2x &+ 4y &&= 9. \end{aligned}$$

Subtracting twice row 1 from row 2 gives

$$0x + 0y = 1.$$

This contradictory row proves inconsistency. In the column picture, $(4, 9)^T$ does not belong to $\operatorname{span}\{(1, 2)^T, (2, 4)^T\}$, the line $v_2 = 2v_1$.

For the right-hand side $(4, 8)^T$, elimination gives $0 = 0$. Only one independent constraint remains, giving

$$x + 2y = 4 \quad \Rightarrow \quad (x, y) = (4, 0) + t(-2, 1),\ t \in \mathbb{R}.$$

A free variable parametrizes an actual direction in the solution set.

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Gaussian elimination

Elimination preserves the solution set through invertible row operations.

1. Exchange two rows.
2. Multiply a row by a nonzero scalar.
3. Add a scalar multiple of one row to another.

Choose a nonzero pivot in the first active column and eliminate entries below it. Continue with the remaining rows and columns. Skip an active column if it contains no available nonzero entry.

For example,

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 3 & 8 & 4 & 24 \\ 0 & 4 & 3 & 17 \end{array}\right]$$

Use row 1 to clear the first entry of row 2, then process the second column. Every operation preserves the equation system's solution set.

Apply an operation to both sides of an equation, rather than changing an isolated entry.

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Pivots and multipliers

To use pivot p at step k to clear position (i, k), the multiplier is

$$\ell_{ik} = \frac{A_{ik}}{p},$$

and the operation is $R_i \leftarrow R_i - \ell_{ik} R_k$.

The preceding example has first multiplier $3/1 = 3$, giving

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 2 & 1 & 0 \\ 0 & 4 & 3 & 17 \end{array}\right].$$

The second pivot is 2 and its multiplier is $4/2 = 2$.

A small nonzero pivot is valid in exact arithmetic but may amplify errors in finite precision. Chapter 11 treats partial pivoting.

Back substitution and rank

Completing the preceding elimination gives

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 17 \end{array}\right].$$

Solve upward from the bottom row.

$$z = 17, \quad 2y + 17 = 0 \Rightarrow y = -17/2,$$
$$x + 2(-17/2) + 17 = 8 \Rightarrow x = 8.$$

There are three nonzero pivots, so this square matrix has rank 3 and the solution is unique. A zero coefficient row with zero right-hand side imposes no new constraint. A nonpivot variable remains free in a consistent system.

Rank is the number of nonzero pivots. Ordinary row echelon form already identifies pivot column indices. A basis for the original column space uses the corresponding original columns, not the changed columns after elimination.

A complete elimination decision procedure

For $A \in \mathbb{F}^{m \times n}$ and $b \in \mathbb{F}^m$, proceed as follows.

1. Fix the field and variable order. In exact arithmetic, choose any nonzero pivot. In floating point, usually choose the largest available absolute value in the current column.
2. Scan columns from left to right. If the current column is zero in all available pivot rows, skip it; its variable is a free variable.
3. Move a nonzero entry to the pivot row, eliminate entries below it in that column, and continue to the right. Each step left-multiplies by an invertible elementary matrix.
4. After reaching row echelon form, look for $[0 \ \cdots \ 0 \mid c]$. A nonzero c makes the system inconsistent. Otherwise it is consistent.
5. To find the nullspace, solve the homogeneous system separately. Keep the echelon coefficient matrix but replace the entire right-hand side by 0. For each free column, set its free variable to 1, all other free variables to 0, and back-substitute the pivot variables. This produces special solutions $As_j = 0$. Keeping the nonhomogeneous right-hand side instead produces solutions of the original system, not nullspace vectors.
6. For a particular solution, set all free variables to 0 and back-substitute using the transformed original right-hand side. The complete solution is

$$x = x_p + \sum_{\text{free } j} t_j s_j.$$

A complete elimination decision procedure (continued)

The chosen field matters. A pivot that is zero in one field cannot be used merely because it is nonzero in another. Apply every row operation to the augmented column when testing consistency.

For $x_1 + 2x_2 = 4$, a particular solution is $x_p = (4, 0)^T$. The homogeneous equation $x_1 + 2x_2 = 0$, with $x_2 = 1$, gives $s = (-2, 1)^T$. Thus $x = (4, 0)^T + t(-2, 1)^T$. Keeping the original right-hand side gives $(2, 1)^T$, whose image is 4, not zero.

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Row operations as matrix products

Adding twice row 1 to row 2 of the identity produces the elementary matrix

$$E = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \qquad EI_2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

Left multiplication acts on rows. Right multiplication acts on columns.

$$EA \text{ changes rows}, \qquad AF \text{ changes columns}.$$

An elementary matrix E is invertible. Its inverse undoes the associated row operation. Here,

$$E^{-1} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}.$$

The side of multiplication determines the operation, so it cannot be interchanged.

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Order of elimination matrices

Applying E_1 first and E_2 second gives

$$E_2(E_1A) = (E_2E_1)A.$$

The rightmost factor acts first. Combining the matrices gives E_2E_1; exchanging them can change the row-operation result.

For a square matrix, suppose elimination uses no row swaps or pivot-row scalings, only subtraction of pivot-row multiples from lower rows. Then

$$E_k \cdots E_2E_1A = U.$$

Consequently,

$$A = (E_k \cdots E_1)^{-1}U = LU.$$

This becomes the LU factorization developed below. The first-column multipliers ℓ_{ik} become entries of L.

The inverse reverses the factor order, $(E_k \cdots E_1)^{-1} = E_1^{-1} \cdots E_k^{-1}$.

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When a row exchange is needed

If a prospective pivot is zero, exchange it with an available nonzero entry below it.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix}.$$

If a final row has the form

$$[0 \ \cdots \ 0 \mid c],$$

with $c \neq 0$, the system is inconsistent. Continue elimination if an exchange can provide a pivot. If all available entries in that column are zero, move to the next column.

Pivots need not lie on the diagonal, particularly for rectangular matrices.

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Matrix multiplication and dimensions

If $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times p}$, then

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}, \qquad AB \in \mathbb{F}^{m \times p}.$$

The inner dimension n must match. The output has the row count of the left matrix and column count of the right matrix.

The product AB is defined only for compatible shapes. Its existence does not imply that BA exists. Even when both exist with the same shape, generally

$$AB \neq BA.$$

A product applies the right transformation first and the left one second. Noncommutativity records the significance of order.

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Column and row combinations

$$AB = A\begin{bmatrix} b_1 & \cdots & b_p \end{bmatrix} = \begin{bmatrix} Ab_1 & \cdots & Ab_p \end{bmatrix}.$$

Each column of AB combines columns of A using the corresponding column of B as coefficients. Alternatively, decompose by rows.

$$AB = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix} B = \begin{bmatrix} a_1^T B \\ \vdots \\ a_m^T B \end{bmatrix}.$$

For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix},$$

Right multiplication by this diagonal sign matrix negates column 2. It does not exchange columns.

Block multiplication requires compatible dimensions at every block interface.

Noncommutativity and distribution

Take

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Hence $AB \neq BA$. The distributive laws still hold.

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC,$$

Associativity $A(BC) = (AB)C$ also holds for compatible sizes. If A is invertible, multiply $AB = AC$ on the left by A^{-1} to obtain $B = C$. Cancellation through a general singular factor is invalid.

Do not transfer commutativity of real multiplication to matrix multiplication.

Why do differently sized products have equal traces?

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$, AB is $m \times m$ while BA is $n \times n$. Both are square, so the trace—the sum of diagonal entries—is defined:

$$\begin{aligned} \operatorname{tr}(AB) &= \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ji} \\ &= \sum_{j=1}^{n} \sum_{i=1}^{m} b_{ji} a_{ij} = \operatorname{tr}(BA). \end{aligned}$$

We rearrange finite sums and commute **scalar** factors; we do not assume $AB = BA$.

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Check the trace with actual matrices

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

$$AB = \begin{bmatrix} 4 & 1 \\ 1 & -3 \end{bmatrix}, \qquad BA = \begin{bmatrix} 2 & 5 & 3 \\ 1 & 2 & 0 \\ 0 & -1 & -3 \end{bmatrix}.$$

Hence $\operatorname{tr}(AB) = 4 - 3 = 1$ and $\operatorname{tr}(BA) = 2 + 2 - 3 = 1$. Equal traces do not make these products equal; their sizes differ.

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Group the same six products in two ways

$a_{ij}\,b_{ji}$	j=1	j=2	j=3	AB diagonal
i=1	1·2 = 2	2·1 = 2	0·0 = 0	4
i=2	0·1 = 0	1·0 = 0	3·(−1) = −3	−3
BA diagonal	2	2	−3	**sum = 1**

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Cyclic rotation is not an arbitrary permutation

When dimensions permit, $\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$: treat BC as one matrix and apply the two-factor identity. Arbitrary exchanges are not generally valid.

For example, set

$$P = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Since $PQ = R$ and $QP = \operatorname{diag}(0, 1)$, $\operatorname{tr}(PQR) = 1$ but $\operatorname{tr}(QPR) = 0$. A cyclic rotation preserves relative order; swapping P, Q does not.

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Trace of a commutator depends on the field

For real or complex $n \times n$ matrices,

$$\operatorname{tr}(AB - BA) = 0 \neq n = \operatorname{tr}(I_n),$$

so $AB - BA = I_n$ is impossible. This uses the fact that n is nonzero in the field.

Over a field of characteristic 2, the previous P, Q satisfy $PQ - QP = I_2$: subtraction equals addition and $\operatorname{tr}(I_2) = 1 + 1 = 0$. The impossibility claim therefore needs a field condition.

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Inverses and one-sided inverses

For $A \in \mathbb{F}^{n \times n}$, if a matrix B satisfies

$$AB = BA = I_n,$$

then A is invertible and $B = A^{-1}$. The definition requires both products to be the identity.

The inverse is unique. If $AB = I$ and $CA = I$, then

$$B = IB = (CA)B = C(AB) = CI = C.$$

so a left inverse and a right inverse must coincide when both exist.

Rectangular matrices require separate one-sided inverse conditions.

If $A \in \mathbb{R}^{m \times n}$, $m \neq n$, it cannot have both $AB = I_m$ and $BA = I_n$. If it has full column rank, $n \leq m$, then $A^T A$ is invertible and

$$A_L = (A^T A)^{-1} A^T, \qquad A_L A = I_n,$$

is a left inverse, not a two-sided inverse. Other left inverses may exist. For full row rank, $m \leq n$, similarly

$$A_R = A^T (A A^T)^{-1}, \qquad A A_R = I_m.$$

Both formulas require their stated dimensions and ranks. Determinants apply only to square matrices. A rank-deficient matrix instead requires a Moore–Penrose pseudoinverse or a least-squares formulation.

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Inverses and one-sided inverses (continued)

Row operations preserve the same unknown vector x. Column operations reorder or mix the unknowns and require the corresponding change of variables. This is why elimination solves equations using row operations.

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Gauss–Jordan inversion

To invert A, augment it with I.

$$[A \mid I],$$

Use row operations until the left side becomes I.

$$[I \mid A^{-1}].$$

For example,

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right].$$

Thus $A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$.

Check $AA^{-1} = I$. If elimination produces a zero row on the left, A is singular.

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Equivalent conditions for invertibility

For square $A \in \mathbb{F}^{n \times n}$, the following statements are equivalent over the specified field.

1. $Ax = 0$ has only the zero solution.
2. A^{-1} exists.
3. The n columns are independent.
4. The columns span $\mathbb{F}^n$.
5. Elimination produces n nonzero pivots.
6. $\det A \neq 0$.

Elimination proves the first five equivalences. A trivial nullspace means no free variables. A square matrix then has n pivots, allowing a unique solution for every right-hand side. Solve for each standard basis right-hand side to assemble the inverse. Conversely, multiplying a homogeneous equation by the inverse proves its solution is zero. Statement 6 uses the determinant developed in Chapter 5. Rectangular matrices do not satisfy all these square-matrix equivalences.

The field matters. An integer matrix may be invertible over the reals but singular modulo a prime.

From elimination to $A = LU$

If elimination needs no row exchanges and all required pivots are nonzero, then

$$A = LU,$$

Here L is unit lower triangular and U is upper triangular. Entry L_{ik} records the multiplier subtracted from row i using row k.

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}.$$

The first multiplier is $4/2 = 2$, giving

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

Check that $LU = A$. Once the factors are available, solving with many right-hand sides requires only forward and back substitution for each one.

Entries of L record elimination multipliers, not the final reduced data.

Row exchanges and $PA = LU$

A zero pivot may require a row exchange. Combining the exchanges into a permutation matrix P gives

$$PA = LU.$$

For an invertible square matrix, suitable row choices always provide nonzero pivots. Left multiplication by P reorders the rows of A.

The factor order matters. $PA = LU$ factors the permuted matrix by elimination without further swaps.

In floating point, nonzero alone may be insufficient. Partial pivoting chooses the largest available absolute value in the column.

For example,

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}, \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

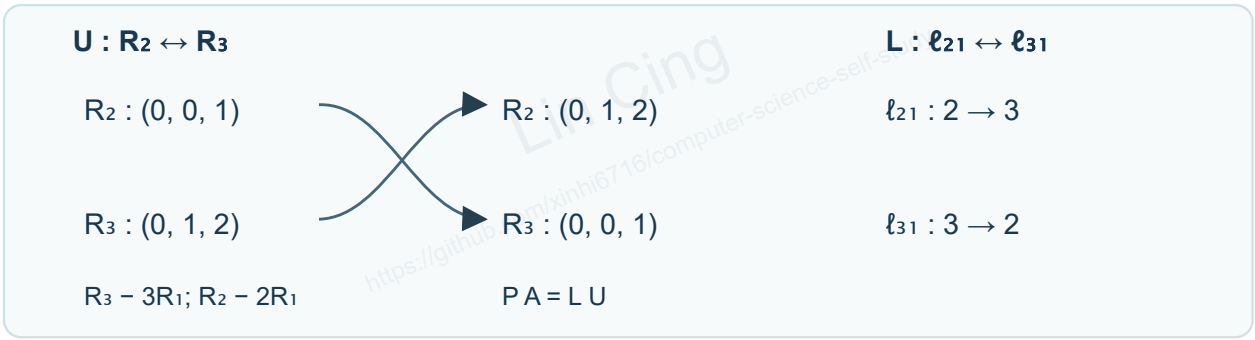
Swapping the rows gives $PA = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} = U$, so $L = I$. Also swap the right-hand side to $Pb = (7, 4)^T$. Back substitution gives $y = 2, x = 1$, and the original system returns $(4, 7)^T$. Swapping only A changes the system. A late swap in a multistep factorization must also swap the already stored multipliers in L; otherwise $PA = LU$ fails even with the correct U.

Row swaps also move previously stored multipliers

Let

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ 6 & 4 & 5 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ 8 \\ 12 \end{bmatrix}.$$

The first pivot is 2, with multipliers $\ell_{21} = 2$ and $\ell_{31} = 3$. After elimination the second pivot position is zero, but the entry below it is 1. Swap rows 2 and 3; this does not mean that the matrix is singular.



Factor and solve the same example

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Direct multiplication gives $LU = PA = \begin{bmatrix} 2 & 1 & 1 \\ 6 & 4 & 5 \\ 4 & 2 & 3 \end{bmatrix}$. Solve $Ly = Pb = (3, 12, 8)^T$ to get $y = (3, 3, 2)^T$; then back-substitute in $Ux = y$ to obtain $x = (1, -1, 2)^T$. Check $Ax = b$ in the original system.

Swapping only the rows of U while leaving the old multipliers in place does not reconstruct PA.

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Invertibility does not eliminate the need for row swaps

For the same matrix, $\det A = -2 \neq 0$, while its leading 2×2 principal submatrix has determinant $2 \cdot 2 - 1 \cdot 4 = 0$. An invertible matrix can therefore encounter a zero pivot when elimination is attempted without row swaps.

One row swap gives $\det P = -1$. Using $PA = LU$, $\det L = 1$, and $\det U = 2$ verifies $\det A = -2$. A swap changes the sign of the determinant and the order of equations, but not the solution set.

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Cost and reuse

For a dense $n \times n$ matrix,

- Gaussian elimination uses $O(n^3)$ arithmetic operations.
- Forward and back substitution use $O(n^2)$.
- Explicit inversion followed by multiplication adds work and is usually unnecessary.

A common strategy factors

$$A \longrightarrow PA = LU$$

once, then substitutes for each new right-hand side. Elimination also tests consistency directly.

Determinants and symbolic inverses can be useful in exact algebra. Numerical computation usually solves through factors.

The leading cost follows by summation. Pivot k updates about $(n-k)^2$ trailing entries with one multiplication and one subtraction each, for a total of

$$2\sum_{k=1}^{n-1}(n-k)^2 = \frac{n(n-1)(2n-1)}{3} = \frac{2}{3}n^3 + O(n^2)$$

Cost and reuse (continued)

floating-point additions and multiplications. Multiplier divisions and right-hand-side updates add $O(n^2)$. A triangular solve processes $n(n-1)/2$ off-diagonal terms and costs $O(n^2)$. This arithmetic model excludes data movement, sparse fill-in, and large-integer bit costs.

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Existence and uniqueness of LU without swaps

An invertible square matrix has an LU elimination with no swaps and nonzero pivots at every step exactly when every leading principal submatrix A_k is invertible. Given such a factorization, triangular structure gives $A_k = L_k U_k$, so

$$\det A_k = \prod_{j=1}^{k} u_{jj} \neq 0.$$

Conversely, induct on elimination steps. The first pivot is $a_{11} = \det A_1 \neq 0$. After the first $k-1$ steps, the determinant of the leading k-by-k block is the product of the previous pivots and the new pivot. Thus

$$u_{kk} = \frac{\det A_k}{\det A_{k-1}} \neq 0, \qquad \det A_0 = 1.$$

and every step is valid. This is stronger than invertibility of A. For example, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is invertible but requires a row swap.

For fixed P, suppose $PA = LU = \widetilde{L}\widetilde{U}$, with both lower-triangular factors having unit diagonal. Then $\widetilde{L}^{-1}L = \widetilde{U}U^{-1}$ is both unit lower triangular and upper triangular, hence is I. The normalized factors are unique for this P, although different pivot rules may choose different permutations.

Transposes and permutations

The transpose is defined by

$$(A^T)_{ij} = A_{ji}.$$

If $A \in \mathbb{F}^{m \times n}$, then $A^T \in \mathbb{F}^{n \times m}$.

A permutation matrix has exactly one 1 in each row and column, with all other entries zero. For example,

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

The product PA permutes rows, while AP permutes columns. Every permutation matrix satisfies $P^T P = I$.

P^T undoes the permutation. Permuting whole equations preserves their solution set.

Inner and outer products

For $x, y \in \mathbb{R}^n$,

$$x^T y \in \mathbb{R}$$

is an inner product with a scalar output.

For $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$,

$$uv^T \in \mathbb{R}^{m \times n},$$

is an outer product. If both vectors are nonzero, column j is $v_j u$, at least one column is nonzero, and every column lies in $\operatorname{span}\{u\}$. The rank is therefore 1. If either vector is zero, every entry is zero and the rank is 0.

For example,

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix},$$

the second row is twice the first, and the rank is 1.

Replacing u by the zero vector gives

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

with no nonzero column spanning the image, hence rank 0. The same holds for $v = 0$. Thus

Inner and outer products (continued)

$$\operatorname{rank}(uv^T) = \begin{cases} 1, & u \neq 0 \text{ and } v \neq 0, \\ 0, & u = 0 \text{ or } v = 0. \end{cases}$$

u ≠ 0 ∧ v ≠ 0

u = (1,2)ᵀ, v = (3,−1)ᵀ

uvᵀ = [[3,−1],[6,−2]]

rank(uvᵀ) = 1

u = 0 ∨ v = 0

u = (0,0)ᵀ, v = (3,−1)ᵀ

uvᵀ = [[0,0],[0,0]]

rank(uvᵀ) = 0

The products $x^T y$ and xy^T have different output types as well as different factor order.

Chapter check

You can now perform the following tasks.

- Explain the same equation $Ax = b$ using rows and columns.
- Use pivots and free variables to distinguish no solution, a unique solution, and infinitely many solutions.
- Construct the elementary matrix for a row operation.
- Explain the difference between left and right multiplication.
- Check the conditions for an inverse.
- Reconstruct L from multipliers and use $PA = LU$ when needed.
- Distinguish $x^T y$ from xy^T.

The exercises require conditions at every step, especially the underlying field and any need for row exchanges.

Chapter 3 describes the pivot structure through column spaces, nullspaces, and bases.

Integral inner products and Gram systems

Prerequisites: inner products, orthogonal projection, and definite integrals.

To extend elimination to function spaces, use the real continuous functions $C([-1, 1], \mathbb{R})$ with

$$\langle f, g \rangle = \int_{-1}^{1} f(t) g(t)\, dt.$$

Continuity ensures integrability. A nonzero continuous function has absolute value bounded below by a positive constant on some interval, so $\int f^2 > 0$. This supplies positive definiteness. An arbitrary pointwise function supported at one point can instead have zero squared integral. In L^2, functions equal almost everywhere must be identified as the same vector.

Take the nonorthogonal basis $p_1 = 1, p_2 = 1 + t$. Each Gram entry is $G_{ij} = \langle p_i, p_j \rangle$.

$$G = \begin{bmatrix} \int 1 & \int (1+t) \\ \int (1+t) & \int (1+t)^2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 8/3 \end{bmatrix}.$$

For $f = 1 + 3t$, the right-hand side is

$$b = \begin{bmatrix} \langle p_1, f \rangle \\ \langle p_2, f \rangle \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Integral inner products and Gram systems (continued)

Finding $f = c_1 p_1 + c_2 p_2$ reduces to the Gram system. The same system gives orthogonal projection when an exact expansion is unavailable.

$$G \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = b.$$

The first equation gives $c_1 + c_2 = 1$. Substitution in the second gives $c_2 = 3$, hence $c_1 = -2$. Check

$$-2(1) + 3(1 + t) = 1 + 3t.$$

Since $\det G = 2(8/3) - 2 \cdot 2 = 4/3 > 0$, the coefficients are unique. In general, $c^T G c = \| \sum_i c_i p_i \|^2$, which is positive for nonzero c when the functions are independent. With dependent functions, G may be singular, but $b_i = \langle p_i, f \rangle$ still belongs to $\mathrm{Col}(G)$. Indeed, $z \in \ker G$ gives $\sum_i z_i p_i = 0$, hence $z^T b = 0$. A projection exists and is unique as a function, although its coefficients may be nonunique. An f outside the span produces a nonzero orthogonal residual, not an inconsistent Gram system.

Use the integral inner product on $C([0, 2\pi], \mathbb{R})$. For the orthogonal basis $\{1, \cos x, \sin x\}$, projection coefficients follow by division. For $f(x) = 1 + x$ on $[0, 2\pi]$,

$$a_0 = \frac{\int_0^{2\pi} (1 + x)\, dx}{2\pi} = 1 + \pi,$$

Integral inner products and Gram systems (continued)

$$a_1 = \frac{\langle \cos x, f \rangle}{\langle \cos x, \cos x \rangle} = 0, \qquad b_1 = \frac{\langle \sin x, f \rangle}{\langle \sin x, \sin x \rangle} = -2.$$

the finite Fourier projection is $f_1(x) = 1 + \pi - 2 \sin x$. The residual $e = x - \pi + 2 \sin x$ is orthogonal to 1, $\cos x$, and $\sin x$. This is a finite-dimensional least-squares projection.

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A parameterized Toeplitz matrix

Prerequisites: determinants, recurrences, and eigenvalues.

For $a \in \mathbb{R}$ and a positive integer n, consider

$$T_n(a) = \begin{bmatrix} a & 1 & & 0 \\ 1 & a & \ddots & \\ & \ddots & \ddots & 1 \\ 0 & & 1 & a \end{bmatrix}.$$

Let $D_n = \det T_n(a)$, with $D_0 = 1$. Expansion along the first row gives

$$D_n = aD_{n-1} - D_{n-2}.$$

The second term comes from the adjacent off-diagonal entries, leaving D_{n-2}. For $n \geq 2$, use the recurrence with $D_0 = 1, D_1 = a$. The first three values are

$$D_1 = a, \qquad D_2 = a^2 - 1, \qquad D_3 = a(a^2 - 1) - a = a(a^2 - 2).$$

An eigenvector derivation is also possible. For $k = 1, \ldots, n$, let $\theta_k = k\pi/(n+1)$, $(v_k)_j = \sin(j\theta_k)$, with boundary values $v_0 = v_{n+1} = 0$. The trigonometric identity gives

$$(v_k)_{j-1} + (v_k)_{j+1} = 2\cos(\theta_k)(v_k)_j,$$

so $T_n(a)v_k = (a + 2\cos\theta_k)v_k$. These nonzero eigenvectors have distinct eigenvalues and are independent. Therefore

A parameterized Toeplitz matrix (continued)

$$D_n(a) = \prod_{k=1}^{n} (a + 2\cos(k\pi/(n+1))).$$

For $n = 3$, the eigenvalues are $a + \sqrt{2}, a, a - \sqrt{2}$, whose product is $a(a^2 - 2)$. Over $\mathbb{R}$, $T_3(a)$ is invertible exactly when $a \notin \{-\sqrt{2}, 0, \sqrt{2}\}$. The parameter conclusion follows from the general determinant structure rather than a single numerical test.

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From elimination to solution sets

Check inconsistent rows and free variables, then use elimination, back substitution, or factorization. Row swaps also affect the right-hand side and previously stored multipliers.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 2.

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